



A butterfly beating its wings in

Chaos

The maths of messes

Some things are easy to predict using a bit of maths. We know exactly where the planets will be in 100 years and how high the tide will rise next Christmas, for instance. Other things are nigh on impossible to predict, like where a pinball will go or what the weather will be like in a week.

The reason is a mathematical phenomenon called *chaos*.

What is chaos?



If something is chaotic, a tiny change in the starting conditions has a huge impact on the final outcome. Pinball makes use of chaos. Each ball you fire takes a different route. Tiny differences in the ball's starting position or the amount you pull the spring become magnified into major changes in direction as it bounces around the table.



Over billions of years, the motion of Earth and the